

Lab 2 - Impinging Flow Analysis

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In this experiment, pressure sensitive paint (PSP) is used to measure how pressure is distributed over a flat surface from an impinging jet flow. PSP allows pressure to be measured at all points simultaneously, rather than at a few points from probes.

Nomenclature

PSP	=	Pressure Sensitive Paint
RGB	=	Red-Green-Blue
P	=	Pressure
P _{ref}	=	Reference Pressure (atmospheric pressure)
I _R	=	Red Intensity
I _G	=	Green Intensity

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Introduction

In this experiment, PSP is used to measure the pressure distribution of a jet flow. “PSP is composed of an oxygen-sensitive fluorescent molecule embedded in an oxygen permeable binder” (MAE 352 Handout 2). The way PSP works is that the more oxygen pressure is applied, the more intense the paint luminesces. The molecules within the PSP are sensitive to oxygen, therefore react when more oxygen is pushed through the permeable membrane. In order for the paint to be measurable, it has to undergo a process called “oxygen-quenching.” The paint is exposed to UV light to excite the molecules by absorbing high energy photons. This leads the molecules to releasing a much lower energy photon to go back down to its ground state. This process is called red-shifting, which emits visible light. The oxygen in the air quenches the amount of visible light depending on the pressure of the oxygen. The resulting brightness is essentially a pressure map of the surface. Higher pressure means more oxygen quenching, and an overall lower brightness.

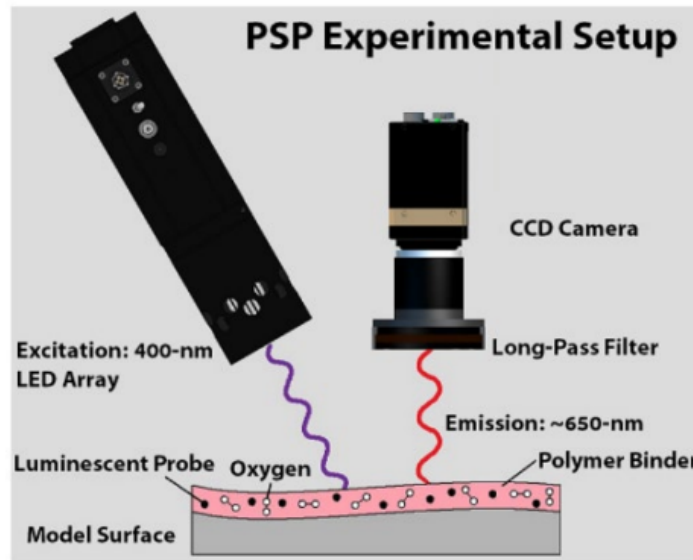


Figure 1: Typical PSP Set-Up

The PSP specifically used in this experiment is what is known as a binary PSP. Only red and green wavelengths are emitted from the paint, which a color camera can separate. There is a pressure sensitive dye, which emits red wavelengths, and a reference dye, which emits green wavelengths. The red dye brightness is based on the oxygen quenching, and the green dye is used to measure luminosity and temperature. The color camera uses a Bayer filter to separate the red and green wavelengths and stores them as pixels. In this experiment, blue wavelengths are not emitted, nor would be useful in any way. A dark image is also taken of the surface. This image is taken with no UV light, and no flow. This measures camera noise, because even with no flow or light, the dark image can still produce nonzero pixel values. Once the pressure wavelengths, and reference wavelengths are stored, the dark image can be subtracted from them and a ratio between the two would provide an accurate pressure map, because it would eliminate uneven lighting, temperature changes, paint thickness variations, and camera sensitivity variations. What is left is a pure measurement of pressure.

Methodology

The first step in this experiment is to set up the PSP system, preferably in a very dark room to eliminate any outside noise. Once the set up is complete, the dark image is taken to get a sense of the camera noise, such as electronic noise, thermal noise, hot pixels, background light, etc.. Once that image is stored, then the UV light is turned on, and another reference image is taken to establish a true baseline. This is because even with no flow, the paint does not glow uniformly. Then the flow is turned on, and images are taken at 20 psi, 25 psi, and 30 psi. The

images are stored as .tif files because .tif files keep the exact data. There is no lossy storage, which just means the data is accurate, unlike JPG or PNG files which will typically approximate large data sets. Once the files are stored, they are read into MATLAB for processing. After the data is processed, the ratio-to-ratio can be calculated.

$$\frac{P}{P_{ref}} = A + B \frac{(I_{R,wind\ off} - I_{R,dark}) / (I_{G,wind\ off} - I_{G,dark})}{(I_{R,wind\ on} - I_{R,dark}) / (I_{G,wind\ on} - I_{G,dark})}$$

Equation I: Ratio-to-Ratio equation to find pressure based on intensity of PSP.

A and B are calibration constants from the Stern-Volmer relationship where $A = 0.14$ and $B = 0.86$. These constants convert the intensity ratios into a real physical quantity, which is pressure in this case. Another important thing to highlight is that the atmospheric pressure was 14.7 psi for all tests.

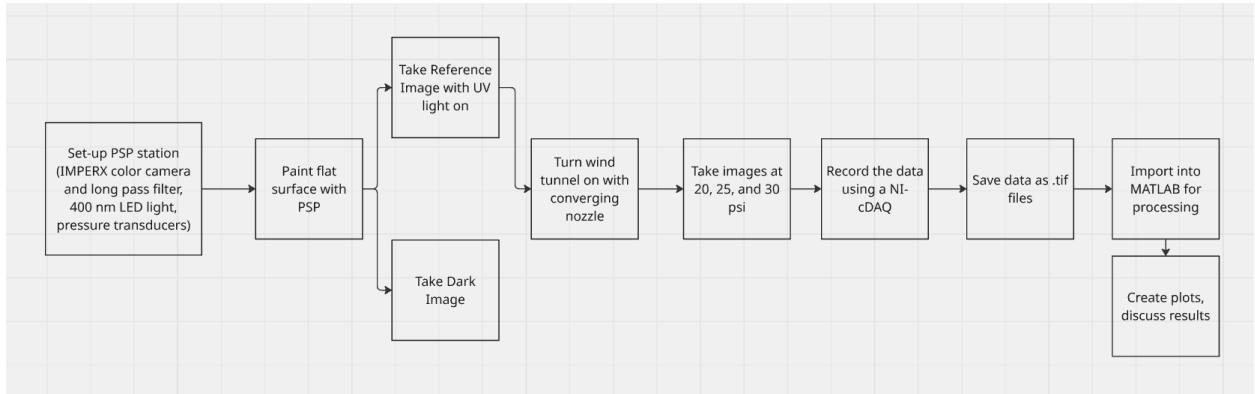


Figure 2: Experimental Flow Diagram

Results

Below show the intensity ratio contours subplotted for all three pressure cases found by using Eq. 1.

Intensity Ratio Contours

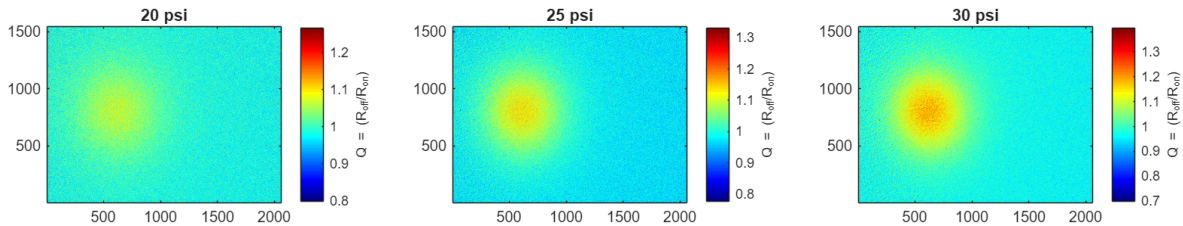


Figure 3: Intensity Ratio Contours

Next shows the pressure field contours for all three pressure cases. This was found by multiplying the ratio-of-ratios by the reference pressure, to get the pressure by itself. Figures 3 and 4 are nearly identical, the only difference is the scale, because all that was done was multiplied by the reference pressure.

Pressure Field Contours

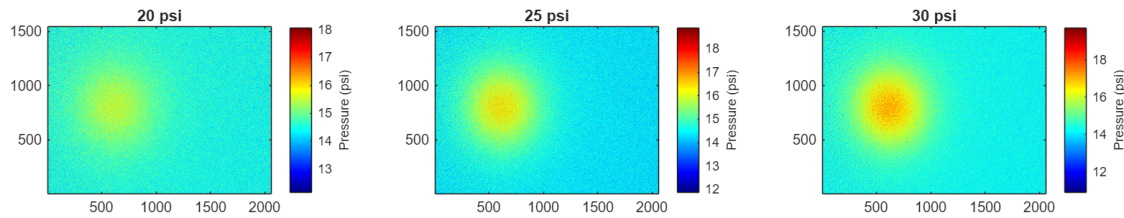


Figure 4: Pressure Field Contours

The next plot shows the horizontal centerline of each pressure case. This essentially turns the 2D pressure plots into a 1D profile. The impact point from the pressure contours is the stagnation pressure, and the pressure decreases away from the center. As shown below, it is the measured pressure from left to right through the stagnation pressure, which is why there is a peak. The second subplot in figure 5 is after Gaussian smoothing, which just makes the data more digestible.

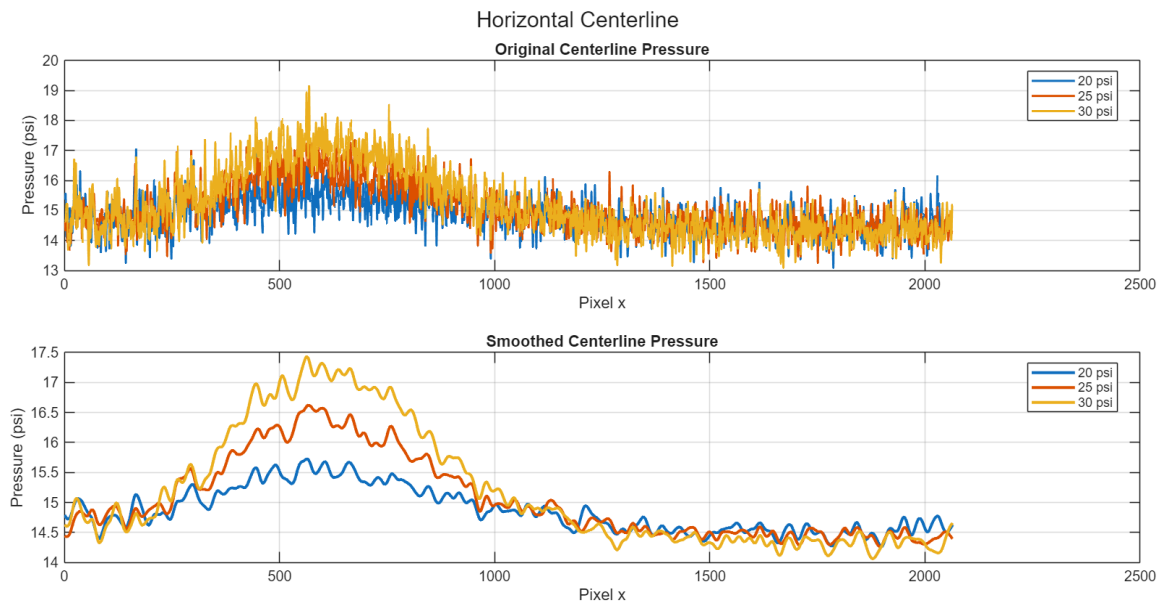


Figure 5: Horizontal Centerline of the Pressure Field Contours

Discussion

The PSP shows the pressure distribution of jet impinging of a flat surface for three cases. For all cases, the pressure was at its maximum in the center of where the flow was impacting the surface, which corresponds to its stagnation pressure. Moving away from the center, the pressure decreases. As the pressure was increased from 20 to

30 psi, there was a noticeable increase in the peak pressure, as well as an overall increase in the contour diameter. This is consistent with the theory that if the jet increases, the pressure will increase as well. The shape of the contours remain the same, which implies that the flow itself is consistent with itself other than its intensity. The centerline plots show a more quantitative analysis of the contours. Each line shows a distinct peak where the pressure reaches its stagnation point. The shapes of the curves themselves vary some, which could be due to multiple reasons such as nozzle alignment, change in lighting, PSP thickness, sensor or camera noise, etc.. Overall, the trendlines remain the same. The original centerline data shows high fluctuations, which inherently proves the existence of noise in some capacity. After smoothing the curves, the fluctuations are limited, but still present. Smoothing does not change the peaks or the overall shape, it just eliminates excess noise.

This experiment is fundamentally successful, in that the higher the jet, the larger the pressure contour. However, due to all of the variable factors, this experiment is extremely difficult to get perfect, and the results may not be accurate. Even within the equation used, the Stern-Volmer constants are just rough approximations of the calibration between intensity and pressure. Other variables include temperature, illumination, camera noise, and many others that may have been mentioned prior.

Overall, this experiment was successful in that PSP can provide pressure distributions based on jet flows.

Appendix

```
%% Lab 2 - Impinging Flow Analysis
% Nathan McIntire
% 2/27/26
clear; clc; close all
% calibration constants
A = 0.14;
B = 0.86;
P_ref = 14.7; % psi (atmospheric pressure)
% Read Images
Dark_image = imread("Dark_Image.tif");
Ref_image = imread("Ref_Image.tif");
Test_image_20psi = imread("Test_Image_20PSI.tif");
Test_image_25psi = imread("Test_Image_25PSI.tif");
Test_image_30psi = imread("Test_Image_30PSI.tif");
%% convert to doubles for calculations and drop Blue
Ref = double(Ref_image(:,:,1:2)); % keep R,G
T20 = double(Test_image_20psi(:,:,1:2));
T25 = double(Test_image_25psi(:,:,1:2));
T30 = double(Test_image_30psi(:,:,1:2));
Dark = double(Dark_image(:,:,1:2));
%% Put test images in a cell array for easy looping (order: 20,25,30)
tests = {Ref, T20, T25, T30};
L = length(tests); % prevents hardcoding
% Loop over tests and store channel images (dark-corrected, non-negative)
for k = 1:L
    img = tests{k}; % H x W x 2 (R,G)
    I_R = img(:,:,1) - Dark(:,:,1); % eliminates noise
    I_G = img(:,:,2) - Dark(:,:,2);
    combine_R(:,:,k) = I_R;
    combine_G(:,:,k) = I_G;
end
% get values from indexing
w_off_red = combine_R(:,:,1);
w_20_red = combine_R(:,:,2);
w_25_red = combine_R(:,:,3);
w_30_red = combine_R(:,:,4);
w_off_green = combine_G(:,:,1);
w_20_green = combine_G(:,:,2);
w_25_green = combine_G(:,:,3);
w_30_green = combine_G(:,:,4);
% compute first ratios
R_off = w_off_red ./ (w_off_green); % constant for each pressure
R_on_20 = w_20_red ./ (w_20_green);
R_on_25 = w_25_red ./ (w_25_green);
R_on_30 = w_30_red ./ (w_30_green);
% Big ratio for each pressure
Big_rat_20 = R_off ./ R_on_20;
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Big_rat_25 = R_off ./ R_on_25;
Big_rat_30 = R_off ./ R_on_30;
P_rat_20 = A + B*Big_rat_20;
P_rat_25 = A + B*Big_rat_25;
P_rat_30 = A + B*Big_rat_30;
%% Intensity Contour figure
figure('name', 'Intensity contour', 'Color','w')
the_big3 = {Big_rat_20, Big_rat_25, Big_rat_30};
plot_titles = {'20 psi','25 psi','30 psi'};
for i = 1:3
    subplot(1,3,i)
    contourf(the_big3{i}, 50, 'LineColor','none')
    axis image
    title(plot_titles{i})
    cb = colorbar;
    cb.Label.String = 'Q = (R_{off}/R_{on})';
end
sgtitle('Intensity Ratio Contours')
colormap jet
%% Convert pressure ratios to absolute pressure fields
P20 = P_ref * P_rat_20;
P25 = P_ref * P_rat_25;
P30 = P_ref * P_rat_30;
% Subplot pressure contours
figure('Name','Pressure Field Contours','Color','w')
P = {P20, P25, P30};
plot_titles = {'20 psi','25 psi','30 psi'};
for i = 1:3
    subplot(1,3,i)
    contourf(P{i}, 50, 'LineColor','none')
    axis image
    title(plot_titles{i})
    cb = colorbar;
    cb.Label.String = 'Pressure (psi)';
end
sgtitle('Pressure Field Contours')
colormap jet
%% Horizontal centerline extraction
rowC = round(size(P20,1)/2); % middle row
x = 1:size(P20,2);
p20 = P20(rowC, :);
p25 = P25(rowC, :);
p30 = P30(rowC, :);
%% Smoothed profiles
span = 40;
p20_s = smoothdata(p20, 'gaussian', span);
p25_s = smoothdata(p25, 'gaussian', span);
p30_s = smoothdata(p30, 'gaussian', span);
%% centerline plots

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```

figure('Name','Horizontal Centerline Pressure Profiles','Color','w')
% ORIGINAL
subplot(2,1,1)
plot(x, p20, 'LineWidth', 1.5); hold on
plot(x, p25, 'LineWidth', 1.5);
plot(x, p30, 'LineWidth', 1.5);
grid on
xlabel('Pixel x')
ylabel('Pressure (psi)')
title('Original Centerline Pressure')
legend('20 psi','25 psi','30 psi','Location','best')
% SMOOTHED
subplot(2,1,2)
plot(x, p20_s, 'LineWidth', 2); hold on
plot(x, p25_s, 'LineWidth', 2);
plot(x, p30_s, 'LineWidth', 2);
grid on
xlabel('Pixel x')
ylabel('Pressure (psi)')
title('Smoothed Centerline Pressure')
legend('20 psi','25 psi','30 psi','Location','best')
sgtitle('Horizontal Centerline')

```